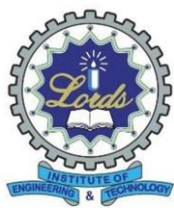
**LORDS INSTITUTE OF ENGINEERING AND TECHNOLOGY****UGC Autonomous**Approved by AICTE | Affiliated to Osmania University
Accredited by NBA | Accredited 'A' grade by NAAC | Certified by ISO.**Department of Science and Humanities****Academic Year: 2026-27****I-B.E. I-SEMESTER****COURSE LEARNING RESOURCES***of***ARTIFICIAL INTELLIGENCE
FOUNDATION****[U26CM101]****[Common for CSE, CSD, CIVIL & MECH]****Prepared****By****Prof. Md Raza Ashraf, Assistant Professor, CSE****Course Faculties:**

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Note: Course Learning Resource is versatile and flexible that cover the entire syllabus of a subject. It is used by students and teachers for learning and assessment purposes only.



Course Outcomes:

Semester No:	1		
Course Title:	Artificial Intelligence Foundation	Course Code:	U26CM101
Course Outcome No.	Description		BTL
AI.CO1	Explain the fundamental concepts, evolution, and impact of Artificial Intelligence.		BTL2 – Understand
AI.CO2	Identify and describe different types of intelligent agents and problem-solving methods used in AI.		BTL2 – Understand
AI.CO3	Understand the basic working principles of Machine Learning.		BTL2 – Understand
AI.CO4	Understand the basic working applications of Machine Learning.		BTL2 – Understand
AI.CO5	Describe major AI application domains such as Natural Language Processing and Computer Vision.		BTL2 – Understand
AI.CO6	Utilize modern AI tools such as ChatGPT, Gamma, and Napkin AI to improve productivity in academic and engineering tasks.		BTL3 – Apply

Course Articulation Matrix:

Mapping of Course Outcomes (CO) with Program Outcomes (PO's) and Program Specific Outcomes (PSO's):

Course Outcomes (CO's)	Program Outcomes (PO)											Program Specific Outcomes (PSO's)	
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2
AI.CO1	2	1	–	–	–	1	1	–	–	–	1	1	–
AI.CO2	2	2	1	–	1	–	–	–	–	–	1	1	2
AI.CO3	2	2	–	–	2	–	–	–	–	–	1	1	2
AI.CO4	2	2	1	–	2	–	–	–	–	–	1	2	2
AI.CO5	2	1	–	–	2	1	1	–	1	–	1	2	1
AI.CO6	1	1	–	–	2	1	1	1	2	1	2	2	1
AI	1.8	1.5	1	–	1.8	1	1	1	1.5	1	1.16	1.5	1.6

Faculty Signature

Course Code	Course Title						Core/Elective
U26CM101	ARTIFICIAL INTELLIGENCE FOUNDATION (Common to CSE, CSD, CIVIL & MECH)						Core
Prerequisite	Contact Hours per Week				CIE	SEE	Credits
	L	T	D	P			
-	1	-	-	-	40	60	01

Course Objectives**The course will introduce the students to**

1. Understand the fundamental concepts, history, and applications of Artificial Intelligence.
2. Learn the basic principles of intelligent agents and problem-solving approaches in AI.
3. Understand the basic concepts of Machine Learning and its role in modern AI systems.
4. Gain knowledge about major AI application areas such as Natural Language Processing, Computer Vision, and intelligent systems.
5. Become familiar with modern AI productivity tools used by engineering students for learning, research, and project development.

Course Outcomes**After successful completion of the course the students will be able to**

1. Explain the fundamental concepts, evolution, and impact of Artificial Intelligence.
2. Identify and describe different types of intelligent agents and problem-solving methods used in AI.
3. Understand the basic working principles of Machine Learning.
4. Understand the basic working applications of Machine Learning.
5. Describe major AI application domains such as Natural Language Processing and Computer Vision.
6. Utilize modern AI tools such as ChatGPT, Gamma, and Napkin AI to improve productivity in academic and engineering tasks.

UNIT I INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Introduction – Definition of Artificial Intelligence – History and Evolution of Artificial Intelligence – Future of Artificial Intelligence – Characteristics of Intelligent Systems – Types of Artificial Intelligence – Applications of Artificial Intelligence in Engineering – Limitations and Challenges of AI – Ethical and Social Issues in AI

UNIT II INTELLIGENT AGENTS AND PROBLEM SOLVING

Intelligent Agents – Structure of Intelligent Agents – Types of Agents – Problem Solving in Artificial Intelligence – Problem Formulation – State Space Representation – Search Strategies (Uninformed & Informed Search)

UNIT III BASICS OF MACHINE LEARNING

Introduction to Machine Learning -Types of Machine Learning – Supervised Learning – Unsupervised Learning – Reinforcement Learning – Training Data and Testing Data – Basic Machine Learning Applications – Examples of Machine Learning in everyday systems.

UNIT IV AI APPLICATION AREAS

Introduction to Natural Language Processing – Text Processing – Chatbots and Virtual Assistants – Introduction to Computer Vision – Image Recognition – Speech Recognition – Recommendation Systems – Applications of AI.

UNIT V AI TOOLS FOR STUDENT PRODUCTIVITY

Introduction to Artificial Intelligence tools for student productivity – Importance of AI tools in learning and time management – AI tools for presentations and documentation such as Gamma – AI tools for visualization and diagram creation such as Napkin AI – AI tools for coding assistance such as GitHub Copilot – AI tools for note organization and documentation such as Notion AI – AI tools for design and presentation creation such as Canva – Applications of AI tools in report writing, presentations, coding assistance and academic productivity.

TEXT BOOKS:

1. S. Russell and P. Norvig, "*Artificial Intelligence: A Modern Approach*", Prentice Hall, Third Edition, 2009.
2. I. Bratko, "*Prolog Programming for Artificial Intelligence*", Addison-Wesley Educational Publishers Inc., Fourth Edition, 2011.
3. P. H. Winston, "*Artificial Intelligence*", Addison-Wesley, Third Edition, 1992.
4. G. F. Luger, "*Artificial Intelligence: Structures and Strategies for Complex Problem Solving*", Pearson Education, Sixth Edition, 2008.

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UNIT-I

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Introduction

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and software capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, recognizing images, and making decisions. AI systems use large amounts of data, algorithms, and computational power to identify patterns and produce useful results.

AI has become an important part of modern life. It is used in smartphones, search engines, online shopping, banking, healthcare, education, transportation, and entertainment. For example, virtual assistants can understand spoken questions, recommendation systems suggest movies or products, and AI-powered medical tools can help doctors analyze patient information.

One major area of AI is **machine learning**, where computers learn from data and improve their performance without being explicitly programmed for every task. Another important area is **deep learning**, which uses artificial neural networks to solve complex problems involving images, speech, and language.

AI offers many benefits, such as increased efficiency, automation, improved decision-making, and faster problem-solving. However, it also raises concerns about privacy, employment, bias, security, and responsible use. Therefore, AI should be developed and used carefully. Overall, Artificial Intelligence is a powerful technology that is transforming society and has significant potential to shape the future.

Keywords : AI → Definition → History → Evolution → Future → Intelligent Systems → Characteristics → Types of AI → Applications → Engineering → Limitations → Challenges → Ethics → Privacy → Security → Bias → Social Impact.

ARTIFICIAL INTELLIGENCE FOUNDATION

UNIT-I

1. INTRODUCTION:

AI is one of the newest fields in science and engineering. Work started in earnest soon after World War II, and the name itself was coined in 1956.

AI currently encompasses a huge variety of subfields, ranging from the general (learning and perception) to the specific, such as playing chess, proving mathematical theorems, writing poetry, driving a car on a crowded street, and diagnosing diseases. AI is relevant to any intellectual task; it is truly a universal field.

1.1 Definitions of AI

Thinking Humanly "The exciting new effort to make computers think ... <i>machines with minds</i> , in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning .. ." (Hellman, 1978)	Thinking Rationally "The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act ." (Winston, 1992)
Acting Humanly "The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	Acting Rationally "Computational Intelligence is the study of the design of intelligent agents." (Poole <i>et al</i> , 1998) "AI ... is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

1.2 History of AI

The idea of "a machine that thinks" dates back to ancient Greece. But since the advent of electronic computing (and relative to some of the topics discussed in this article) important events and milestones in the evolution of AI include the following:

1950

Alan Turing publishes *Computing Machinery and Intelligence*. In this paper, Turing, famous for breaking the German ENIGMA code during WWII and often referred to as the "father of computer science" asks the following question: "Can machines think?"

From there, he offers a test, now famously known as the "Turing Test," where a human interrogator would try to distinguish between a computer and human text response. While this test has undergone much scrutiny since it was published, it remains an important part of the history of AI, and an ongoing concept within philosophy as it uses ideas around linguistics.

1956

John McCarthy coins the term "artificial intelligence" at the first-ever AI conference at Dartmouth College. (McCarthy went on to invent the Lisp language.) Later that year, Allen Newell, J.C. Shaw and Herbert Simon create the Logic Theorist, the first-ever running AI computer program.

1967

Frank Rosenblatt builds the Mark 1 Perceptron, the first computer based on a neural network that "learned" through trial and error. Just a year later, Marvin Minsky and Seymour Papert publish a book titled Perceptrons, which becomes both the landmark work on neural networks and, at least for a while, an argument against future neural network research initiatives.

1980

Neural networks, which use a backpropagation algorithm to train itself, became widely used in AI applications.

1995

Stuart Russell and Peter Norvig publish Artificial Intelligence: A Modern Approach, which has become one of the leading textbooks in the study of AI. In it, they delve into four potential goals or definitions of AI, which differentiates computer systems based on rationality and thinking versus acting.

1997

IBM's Deep Blue beats then world chess champion Garry Kasparov, in a chess match (and rematch).

2004

John McCarthy writes a paper, What Is Artificial Intelligence?, and proposes an often-cited definition of AI. By this time, the era of big data and cloud computing is underway, enabling organizations to manage ever-larger data estates, which will one day be used to train AI models.

2011

IBM Watson® beats champions Ken Jennings and Brad Rutter at Jeopardy! Also, around this time, data science begins to emerge as a popular discipline.

2015

Baidu's Minwa supercomputer uses a special deep neural network called a convolutional neural network to identify and categorize images with a higher rate of accuracy than the average human.

2016

DeepMind's AlphaGo program, powered by a deep neural network, beats Lee Sodol, the world champion Go player, in a five-game match. The victory is significant given the huge number of possible moves as the game progresses (over 14.5 trillion after just four moves). Later, Google purchased DeepMind for a reported USD 400 million.

2022

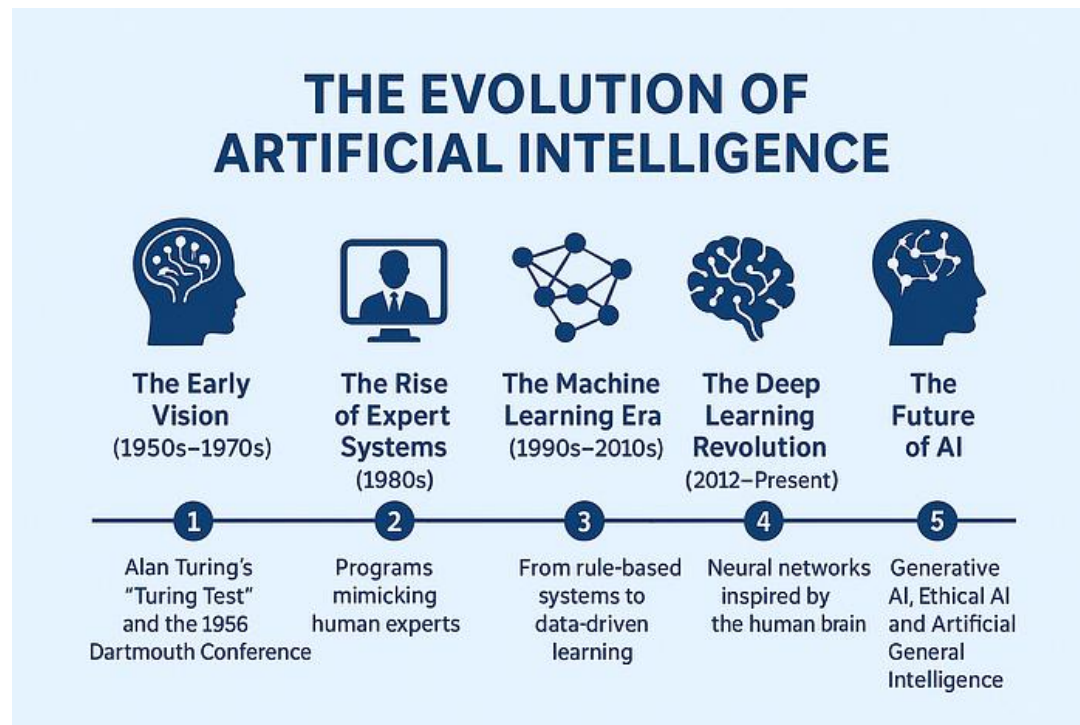
A rise in large language models or LLMs, such as OpenAI's ChatGPT, creates an enormous change in performance of AI and its potential to drive enterprise value. With these new generative AI practices, deep-learning models can be pretrained on large amounts of data.

2024

The latest AI trends point to a continuing AI renaissance. Multimodal models that can take multiple types of data as input are providing richer, more robust experiences. These models bring together computer vision image recognition and NLP speech recognition capabilities. Smaller models are also making strides in an age of diminishing returns with massive models with large parameter counts.

1.3 The Evolution of Artificial Intelligence: From Concept to Reality

Artificial Intelligence (AI) is no longer just a buzzword or a futuristic concept — it is shaping the way we live, work, and interact every day. From voice assistants on our smartphones to intelligent automation in enterprises, AI has evolved dramatically over the last few decades. But how did we get here? Let's take a journey through the evolution of AI and understand its past, present, and future



Evolution of AI

1.3.1. The Early Vision (1950s — 1970s)

The idea of creating machines that could “think” like humans began long before we had the technology to make it real.

- **Alan Turing**, in his famous 1950 paper, proposed the “Turing Test” to evaluate a machine’s ability to exhibit intelligent behavior.
- In the **1956 Dartmouth Conference**, the term *Artificial Intelligence* was officially coined, marking the birth of AI as a research field.
- Early AI systems focused on symbolic reasoning and problem-solving, leading to programs that could play chess or solve logic puzzles.

However, due to limited computing power and data, progress was slow, leading to what we now call the **first AI winter**.

1.3.2. The Rise of Expert Systems (1980s)

In the 1980s, AI research shifted towards **expert systems** — programs designed to mimic the decision-making of human experts.

- These systems were widely used in medicine, engineering, and business.
- A famous example is **MYCIN**, an expert system for diagnosing infections.

This was the first time AI started to see commercial applications, though the technology was still constrained by expensive hardware and limited adaptability.

1.3.3. The Machine Learning Era (1990s — 2010s)

The explosion of computing power and availability of digital data gave rise to **Machine Learning (ML)** — a shift from rule-based systems to algorithms that could learn from data.

- **1997:** IBM's **Deep Blue** defeated world chess champion Garry Kasparov, proving machines could surpass human intelligence in specific tasks.
- **2000s:** Growth of the internet produced massive amounts of data, fueling ML algorithms like decision trees, support vector machines, and later, neural networks.
- **2011:** IBM's **Watson** won *Jeopardy!* by processing natural language at scale, highlighting AI's ability to understand and respond to human queries.

1.3.4. The Deep Learning Revolution (2012 — Present)

The real breakthrough came with **Deep Learning**, powered by advanced neural networks inspired by the human brain.

- **2012:** A deep neural network trained on ImageNet achieved record-breaking accuracy in image recognition, igniting a new wave of AI research.
- AI became embedded in daily life through **voice assistants** like Siri, Alexa, and Google Assistant, **autonomous vehicles**, and **real-time translation tools**.
- Companies like Google, Amazon, and OpenAI invested heavily in AI research, making it accessible to businesses and consumers worldwide.

Generative AI (like ChatGPT, DALL·E, and MidJourney) now creates text, images, and even music — pushing the boundaries of creativity.

This period marked the beginning of AI in everyday applications — from email spam filters to recommendation engines on e-commerce platforms.

1.3.5. The Future of AI: Where Are We Heading?

As we look ahead, AI's evolution is just beginning. Emerging trends include:

- **Generative AI for business and creativity** — transforming industries from education to design.
- **Ethical AI** — addressing concerns of bias, transparency, and responsible use.
- **AI in healthcare** — from drug discovery to personalized treatment plans.
- **Artificial General Intelligence (AGI)** — the ambitious goal of creating machines that can perform any intellectual task humans can.

1.4 The Future of Artificial Intelligence

Artificial intelligence is the force behind continuing breakthroughs. It even goes beyond what humans ever thought would be achievable. According to a report, the AI market will go from \$150.2 billion in

2023 to \$1,345.2 billion by 2030. So, what is the future of AI? Below is a glimpse of what the future of AI could look like.

1.4.1 Job Creation

AI may affect 40% of jobs globally and impact economies. While it will significantly disrupt the job market by automating many existing roles, it will also create new jobs as it transforms and generates new industries. In advanced economies, most jobs may benefit from AI integration and productivity enhancement.

Roles that focus on human creativity, social-emotional skills, and complex management may remain untouched. New jobs will emerge, such as AI specialists, robotics engineers, and user experience designers for AI products. Most jobs will change as automation shifts the mix of skills needed towards more STEM, computer programming, and critical thinking abilities. The solution is to take proactive measures to retrain and upskill to minimize displacement and maximize job creation potential.

1.4.2 Industry Changes

AI will drive widespread transformations in industries as it becomes more integrated into business operations. By automating repetitive tasks, AI will also boost productivity and assist humans with enhanced data analytics, dynamic decision-making, and personalized recommendations.

The gains in efficiency and productivity will cause a ripple effect and spur the emergence of new business and operating models. Below are some of the primary industries that AI will impact:

1.4.3 Healthcare

AI will help diagnose diseases, personalize treatment plans, monitor patients remotely, reduce dosage errors, and more. For instance, 70% of insulin users currently don't follow their prescriptions. Detecting such errors can help improve healthcare.

1.4.4 Education

AI will revolutionize education by enabling personalized and tailored learning experiences for students based on their abilities. Customized learning will provide opportunities for both high-performing and struggling students.

1.4.5 Finance

AI algorithms are already applicable to stock trading and investment fund management. Because they can analyze massive amounts of data, AI systems may outperform humans in financial decision-making. The finance industry will benefit from risk and compliance management as AI helps prevent financial disasters.

1.4.6 Military and Cybersecurity

AI will enable the development of autonomous weapon systems and robots for military purposes. It will also help strengthen cybersecurity through technologies like predictive threat analysis and prevention systems.

1.4.7 Transportation

Self-driving vehicles are one of the wildest technological realizations. AI will also help manage traffic and predict congestion to optimize travel time, monitor vehicle performance, and ensure timely repair and maintenance.

1.4.8 Advertising

AI will enable personalized ads tailored to individual users based on their specific interests, behaviors, locations, and profiles, improving targeting and effectiveness. It's no wonder that over 80% of businesses are testing AI in their marketing channels.

1.4.9 Communication

With artificial intelligence, future communication barriers will break and improve thanks to AI's ability to understand human meanings beyond words and enhance accessibility for those with disabilities. Advanced AI systems that can hold human-level conversations across diverse topics and contexts will usher in new modes of seamless collaboration between diverse groups.

Communication may become more personalized through AI's proficiency in nuanced natural language generation, recognition of individual styles, and contextual awareness. Meanwhile, virtual worlds and AI-driven media will expand the possibilities for shared creative expression. AI is poised to revolutionize communication by making it more inclusive, effective, and personalized, enhancing both interpersonal interactions and global connectivity.

1.5 Characteristics of Intelligent Systems

The research surfaced 13 key characteristics that are critical markers for successful intelligent systems. It also highlighted the importance of investing in the right elements at the right time, or "blueprinting." In the case of building intelligent systems, three concepts are key: prepare the right infrastructure first, next work on foundational needs, and then address longer-term capabilities.

The 13 key characteristics are:

- Ability to simulate and emulate in near real time
- Automated learning and machine learning functionalities
- Digital feedback loops that influence product development
- Action based on sensory data and algorithms
- Customized device experience in the cloud
- True compute on the far edge
- Adapting tasks based on reprogramming via cloud
- Ability to predict stresses and failures
- Detection and resolution of events
- Total automation
- Near-real-time, seamless connections across multiple ecosystems
- Real-time collaborative workflow platform
- Experimenting as a learning system

1.6 Types of AI

1.6.1. Artificial Narrow AI

Artificial Narrow Intelligence, also known as Weak AI (what we refer to as Narrow AI), is the only type of AI that exists today. Any other form of AI is theoretical. It can be trained to perform a single or narrow task, often far faster and better than a human mind can.

However, it can't perform outside of its defined task. Instead, it targets a single subset of cognitive abilities and advances in that spectrum. Siri, Amazon's Alexa and IBM Watson® are examples of Narrow AI. Even OpenAI's ChatGPT is considered a form of Narrow AI because it's limited to the single task of text-based chat.

1.6.2 General AI

Artificial General Intelligence (AGI), also known as Strong AI, is today nothing more than a theoretical concept. AGI can use previous learnings and skills to accomplish new tasks in a different context without the need for human beings to train the underlying models. This ability allows AGI to learn and perform any intellectual task that a human being can.

1.6.3. Super AI

Super AI is commonly referred to as artificial superintelligence and, like AGI, is strictly theoretical. If ever realized, Super AI would think, reason, learn, make judgements and possess cognitive abilities that surpass those of human beings.

The applications possessing Super AI capabilities will have evolved beyond the point of understanding human sentiments and experiences to feel emotions, have needs and possess beliefs and desires of their own.

At its core, the real world applications of artificial intelligence include how AI can enhance design, efficiency, safety, and decision-making across many different engineering fields. AI has been known to improve products from the design phase to the prediction of failures, drive innovation, boost engineers ideas and decision-making abilities.

Research has already shown that AI is changing the meaning of engineering. Predictive maintenance, design and computer-aided design optimization, robotics, civil infrastructure, energy systems, and autonomous vehicles are all areas of fundamental transformation.

1.7 Applications of AI in Engineering

1.7.1 MANUFACTURING & INDUSTRIAL ENGINEERING

1.7.1.1 PREDICTIVE MAINTENANCE & SMART MANUFACTURING

AI-powered sensors monitor machine health and predict failures before they happen. Industrial IoT systems, for example, can reduce breakdowns by up to 70% and cut maintenance costs by 30%. In workshops like Ludhiana's CICU event, experts highlighted AI-driven quality control and predictive maintenance as game changers.

1.7.1.2 ROBOTICS & AUTOMATION

Smart factories incorporate AI robots that learn and adapt in assembly lines, such as BMW's iFactory or elements of General Motors' predictive AI systems, which can optimize task performance, detect defects, and assist human operators.

1.7.1.3 GENERATIVE DESIGN

AI tools can produce thousands of design options based on your requirements. General Motors used this to redesign a seat bracket, resulting in a version that was 40% lighter and 20% stronger.

1.7.2 AUTOMOTIVE & AEROSPACE ENGINEERING

1.7.2.1 SMART DESIGN & SIMULATION

Ford uses AI agents and NVIDIA GPU software to rapidly generate 3D models, perform stress and aerodynamics analysis in seconds (instead of hours). MIT's DrivAerNet++ tool analyzes thousands of car designs to identify the most aerodynamic, cutting time and cost.

1.7.2.2 AUTONOMOUS SYSTEMS

From self-driving vehicles to drones surveying construction sites, AI is powering autonomous systems in engineering. Boston Dynamics' Spot robot, for example, autonomously inspects sites and collects data.

1.7.3 ENERGY & POWER ENGINEERING

1.7.3.1 SMART GRIDS & ENERGY OPTIMIZATION

AI helps make renewable energy systems smarter and more efficient. It accurately forecasts energy use, balances loads in real time, and improves smart grid operations, reducing waste and preventing overloads.

Meanwhile, Microsoft and Idaho National Laboratory are working together to speed up nuclear plant permitting. They're using AI to automatically draft engineering and safety reports from historical data, so regulators and engineers can review and edit and not write from scratch, cutting down weeks or even months.

1.7.3.2 SURROGATE MODELING & DIGITAL TWINS

Engineers use AI to create digital twins, virtual replicas of systems, to simulate and optimize energy performance, as seen in recent digital twin research.

1.7.4 HEALTHCARE & BIOMEDICAL ENGINEERING

1.7.4.1 MEDICAL IMAGING & DIAGNOSTICS

In healthcare, AI's engineering applications extend to diagnostic systems. For example, AI helps detect heart conditions with high accuracy, using data from wearables and patient records.

1.7.4.2 DEVICE DESIGN AND TELEMEDICINE

The applications are numerous: AI-assisted surgery robots, smart prosthetics, and remote monitoring equipment. Application engineers construct these devices by integrating sensors and data with AI to create higher-level intelligence, and they build telemedicine capabilities into the devices to enhance access.

1.7.5 CIVIL & INFRASTRUCTURE ENGINEERING

1.7.5.1 OPTIMIZING STRUCTURES & MATERIALS

AI assists with the design of roads, bridges, and buildings, predicting hazards and the optimization of layout. Civil engineers use neural networks, fuzzy logic, and genetic algorithms for solving complex real-world problems.

1.7.5.2 DIGITAL INSPECTION & QUALITY CONTROL

AI-powered image recognition detects cracks or anomalies in infrastructure. This improves monitoring and ensures safety.

1.8 Challenges and limitations

The limitations of artificial intelligence (AI) encompass various aspects that pose challenges to its development, widespread adoption and application. Some of the key challenges and limitations include:

- Ethical Concerns and Bias
- Safety and Security
- Data Quality and Privacy
- Limited Contextual Understanding
- Interpretability and Explainability
- Lack of Emotional Intelligence and Human Intuition
- Workforce Displacement and Economic Impact

1.8.1 Ethical Concerns and Bias

As AI systems become more prevalent, ethical concerns and biases have garnered significant attention. AI algorithms/systems are trained on vast datasets that can unintentionally perpetuate biases present in the data, leading to discriminatory outcomes. These biases can result from historical data or reflect societal biases. Organizations and researchers are actively working to mitigate bias and design AI systems that promote inclusivity and equity.

Examples of AI ethical concerns and bias include:

Hiring and Recruitment Bias: AI algorithms used in hiring and recruitment processes can inadvertently perpetuate biases present in historical data. For instance, if historical data shows a gender or racial bias in hiring decisions, AI systems trained on that data may inadvertently learn and replicate those biases, Reinforcing inequality in employment opportunities.

Financial Discrimination: AI systems used in financial institutions for credit scoring and loan approvals can unintentionally discriminate against certain demographics. If historical lending practices exhibited biases, AI algorithms trained on that data may perpetuate discriminatory practices, denying loans or offering unfavorable terms to certain groups.

Criminal Justice System Bias: AI algorithms utilized in the criminal justice system, such as predictive policing or sentencing algorithms, can exhibit biases that disproportionately impact certain communities. These biases can result in unfair targeting or sentencing based on factors such as race, socio-economic status, or geographical location.

Addressing these ethical concerns and biases requires transparency, accountability, and responsible AI development practices. Measures such as diverse and representative training data, algorithmic audits,

ongoing monitoring, and bias mitigation techniques can help solve these issues and ensure the fair and ethical use of AI technologies.

1.8.2 Safety and Security

Ensuring the safety and security of AI systems is a significant challenge. Vulnerabilities and potential risks, such as adversarial attacks or system failures, need to be addressed to prevent unintended consequences of AI technology. Malicious actors intentionally manipulate input data to deceive or trick the AI algorithms, leading to potentially harmful consequences in sectors like cybersecurity, autonomous vehicles, and fraud detection. Ensuring the robustness and resilience of AI models against such attacks is an ongoing challenge that demands continuous innovation, rigorous testing, and the development of countermeasures.

Examples of AI safety and security challenges include:

Adversarial Attacks: Adversarial attacks involve deliberately manipulating input data to deceive AI systems. For example, adding imperceptible perturbations to images can trick image recognition systems into misclassifying objects. Adversarial attacks pose security risks in various domains, including autonomous vehicles, cybersecurity, and facial recognition systems.

System Failure and Unreliability: AI systems can experience unexpected failures or errors that may have severe consequences. For instance, autonomous vehicles relying on AI algorithms for decision-making could encounter situations where the system fails to respond appropriately, leading to accidents or safety risks.

Misuse of AI Technology: AI can be misused for malicious purposes, such as developing autonomous weapons or conducting surveillance without appropriate oversight. The potential misuse of AI raises ethical and security concerns, necessitating the establishment of regulations and guidelines to prevent abuse.

Dependence on Internet Connectivity: Many AI systems rely on continuous Internet connectivity to function properly. Disruptions in connectivity or denial-of-service attacks can impact the availability and reliability of AI services, especially cloud-based AI applications.

Safety and security challenges require rigorous testing, robust system design, continuous monitoring, and adherence to best practices. Collaboration between AI developers, cybersecurity experts, policymakers, and regulatory bodies is essential to establish frameworks that promote safety, security, and responsible AI deployment.

1.8.3 Data Quality and Privacy

AI algorithms heavily rely on large volumes of high-quality data for training and decision-making. However, acquiring and maintaining such data can be a daunting task. Issues such as data scarcity, incompleteness, and inconsistency can hinder the performance of AI systems. Additionally, concerns regarding data privacy and security arise when handling sensitive information. Striking a delicate balance between data utility and privacy is essential to build trust and comply with regulations, while still extracting valuable insights from data.

Examples of AI data quality and privacy challenges include:

Biased Training Data: AI models heavily rely on training data to learn patterns and make predictions. If the training data is biased or unrepresentative of the real-world population, the AI system may

replicate or amplify those biases, leading to unfair outcomes. For example, if historical employment data is biased towards a certain demographic, an AI hiring system trained on that data may perpetuate the bias in candidate selection.

Insufficient or Incomplete Data: AI models require large amounts of high-quality data to perform effectively. However, in some domains or emerging fields, obtaining sufficient and comprehensive data can be challenging. Limited or incomplete data can hinder the AI system's ability to generalize and make accurate predictions.

Data Privacy and Security Breaches: AI systems often handle sensitive data. Inadequate data protection measures or vulnerabilities in the system can lead to data breaches, compromising individuals' privacy rights or exposing confidential information. Protecting data privacy is crucial to maintain trust and comply with legal and ethical obligations.

Data Ownership and Consent: AI systems may rely on user-generated data or data collected through various sources. Ensuring proper data ownership, consent, and transparency in data collection and usage is essential to protect individual rights and maintain user trust. Obtaining informed consent and clear data usage policies are important for maintaining data privacy.

To tackle these challenges related to data quality and privacy, it is necessary to establish strong data governance practices, ensure the availability of diverse and inclusive datasets, embrace privacy-enhancing technologies, and foster a culture of transparency and accountability in the collection and utilization of data. It is necessary to respect individuals' data privacy rights, comply with data protection regulations, and uphold ethical guidelines to ensure the responsible development and deployment of AI systems.

1.8.4 Limited Contextual Understanding

Although AI has made remarkable strides in natural language processing and computer vision, it still struggles with contextual understanding and common-sense reasoning. AI models may misinterpret nuanced language, sarcasm, or cultural references, resulting in inaccurate results or inappropriate actions. Advancing contextual understanding requires the development of sophisticated language models and context-aware AI systems that can grasp the intricacies of human communication.

Examples of limited contextual understanding in AI include:

Ambiguity in Natural Language: AI systems can struggle to disambiguate words or phrases with multiple meanings, leading to confusion in understanding the intended context. This can result in misunderstandings or incorrect inferences in natural language processing tasks, such as machine translation or sentiment analysis.

Human-like Reasoning: AI systems typically lack the ability to reason and understand complex scenarios in the same way humans do. They may struggle to apply common sense or logical reasoning to solve problems or make decisions, limiting their ability to perform tasks that require deep contextual understanding.

Language Processing: AI language models often struggle with understanding and interpreting the context of language, leading to misinterpretations or inaccurate responses. For instance, they may struggle with sarcasm, idioms, or cultural references, resulting in incorrect or inappropriate outputs.

Resolving these limitations requires advancements in natural language processing, machine comprehension, multimodal learning, and context-aware AI systems. Ongoing research focuses on

developing AI models that can better understand and reason within complex contexts, bridging the gap between human-level contextual understanding and AI capabilities.

1.8.5 Interpretability and Explainability

Many AI models, particularly deep learning algorithms, operate as "black boxes." They provide outputs without clear explanations or insights into their decision-making process. This lack of interpretability and explainability can be problematic, particularly in critical applications/domains such as healthcare and finance where transparency and accountability are key. Researchers are actively exploring methods to enhance interpretability and provide explanations for AI model outputs, aiming to build trust and facilitate human-AI collaboration.

Examples of AI interpretability and explainability challenges include:

Deep Learning Models: Deep learning algorithms, such as deep neural networks, are often regarded as black boxes due to their complex and opaque nature. Understanding how these models arrive at specific predictions can be challenging, making it difficult to explain their outputs to users or stakeholders.

Medical Diagnosis: In healthcare, interpretability and explainability are vital for gaining trust and acceptance from medical professionals. AI systems used for medical diagnoses, such as image-based diagnosis or clinical decision support, need to provide explanations for their conclusions to assist healthcare providers in understanding the reasoning behind the recommendations.

Financial Risk Assessment: In the finance industry, interpretability and explainability are vital for risk assessment models. Banks and financial institutions need to understand how AI systems arrive at risk scores or credit decisions to ensure fair lending practices and regulatory compliance.

Researchers and practitioners are actively working on developing interpretability and explainability techniques for AI systems. Methods such as model-agnostic approaches, rule extraction, attention mechanisms, and local feature importance analysis aim to provide insights into how AI models arrive at their predictions and decisions, enhancing transparency and trust in AI technology.

1.8.6 Lack of Emotional Intelligence and Human Intuition

Human decision-making often involves intuition, gut feelings, and instinctive judgments based on subtle cues or prior experiences. AI systems rely on data and algorithms, lacking the intuitive leaps that humans can make when faced with ambiguous or uncertain situations. This limitation hinders its ability to understand and respond to complex human emotions, interpersonal dynamics, and subjective aspects of decision-making that require empathy and intuition.

Examples of AI's lack of emotional intelligence and human intuition include:

Customer Service Chatbots: AI-powered chatbots often struggle to understand and respond appropriately to customer emotions and complex inquiries. They may provide generic or robotic responses that fail to address the underlying emotional needs of customers, leading to frustration and dissatisfaction.

Personalized Recommendations: While AI recommendation systems can analyze user preferences and historical data to suggest products, services, or content, they may struggle to capture the deeper emotional and subjective aspects of individual tastes and preferences. This can result in

recommendations that fail to resonate with users on an emotional level or overlook serendipitous discoveries.

Humanoid Robots: Despite advancements in robotics, humanoid AI robots still struggle to display genuine emotions or understand complex human emotional cues. While they can mimic certain emotional expressions, their responses often lack the depth and authenticity that human intuition and emotional intelligence provide.

Creative Fields: In creative domains such as art, music, or writing, AI algorithms can generate content based on patterns and existing examples. However, they may lack the emotional depth, personal experiences, and intuition that human creators bring to their work, limiting their ability to produce truly original and emotionally resonant creations.

To overcome the deficiency of emotional intelligence and human intuition in AI, it is essential to foster collaboration across different disciplines, prioritize user-centric design, and adhere to ethical development principles. By integrating emotional aspects into AI systems and recognizing their wider societal implications, we can work towards the creation of AI technologies that possess a deeper understanding of human emotions and experiences, enabling more empathetic and responsive interactions.

1.8.7 Workforce Displacement and Economic Impact

The widespread adoption of AI technologies can lead to workforce displacement and job market disruptions. It poses challenges for reskilling and upskilling the workforce to adapt to changing job requirements and societal implications. Navigating the economic impact and ensuring a just transition are crucial considerations.

Examples of AI workforce displacement and economic impact include:

Retail Sector: The rise of e-commerce and AI-powered automation in retail has led to the closure of physical stores and a shift towards online shopping. This displacement of retail workers has been particularly significant in roles such as cashiers, inventory management, and customer service, as AI-driven technologies replace human labor with automated systems.

Manufacturing Industry: The adoption of AI-powered robots and automation in manufacturing processes has resulted in job losses for workers involved in repetitive and manual tasks. As AI-driven machines become more efficient and cost-effective, they can replace human workers in areas such as assembly line production and quality control.

Transportation and Delivery Services: The development of autonomous vehicles and drones has the potential to disrupt the transportation and delivery sectors. As self-driving cars and drones become more prevalent, there may be a decrease in demand for truck drivers, delivery personnel, and related occupations.

Customer Support and Call Centers: AI chatbots and virtual assistants are increasingly being used to handle customer inquiries and support services. While these AI systems provide faster response times and cost savings for companies, they can lead to job displacement for customer support representatives and call center agents.

Society can mitigate the negative consequences of AI-driven workforce displacement and promote economic resilience by equipping individuals with the skills, resources, and support necessary to thrive in the changing job landscape.

1.9 Ethical And Societal Implications

AI offers many benefits but poses complex significant risks to ethics and society without responsible applications. AI systems take on more autonomous decision-making roles at the risk of algorithmic bias, lack of transparency, privacy and data security, and job displacement.

Delegating decision-making responsibilities to AI threatens to exacerbate discrimination, economic inequality, and erosion of accountability. Interdisciplinary and ethical guidelines that promote fairness, accountability, and non-discrimination will help to address these issues. As AI gains autonomy, it will be necessary to have frameworks that will hold organizations accountable for transparency.

Without adequate precautions and governance, the growing use of AI could undermine principles of justice, transparency, and the greater public interest by automating and amplifying existing societal biases and inequities.

Exercise

Unit I – Fill in the Blanks

1. Artificial Intelligence is a branch of _____ that deals with creating intelligent machines.
2. The term “Artificial Intelligence” was coined by _____ in 1956.
3. AI systems are designed to perform tasks that normally require _____ intelligence.
4. An AI system that is designed to perform a specific task is called _____ AI.
5. The ability of an intelligent system to improve its performance from experience is called _____.
6. _____ is one of the important applications of AI in engineering for detecting equipment failures before they occur.
7. AI systems may produce unfair results because of _____ in training data.
8. _____ is an important ethical issue related to the collection and use of personal data by AI systems.
9. Robots that can sense their environment and perform tasks using AI are examples of _____ systems.
10. One major social concern associated with AI is _____ displacement due to automation.

Unit I – Multiple Choice Questions (MCQs)

1. **What does AI stand for?**
 - a) Automated Information
 - b) Artificial Intelligence
 - c) Advanced Information
 - d) Automated Intelligence
2. **Who coined the term “Artificial Intelligence”?**
 - a) Alan Turing
 - b) John McCarthy
 - c) Marvin Minsky
 - d) Herbert Simon
3. **Which of the following is a characteristic of an intelligent system?**
 - a) Learning
 - b) Reasoning

- c) Problem-solving
 - d) All of the above
4. **AI designed to perform a specific task is known as:**
- a) General AI
 - b) Super AI
 - c) Narrow AI
 - d) Strong AI
5. **Which of the following is an application of AI in engineering?**
- a) Predictive maintenance
 - b) Fault detection
 - c) Design optimization
 - d) All of the above
6. **Which AI technology enables machines to learn from data?**
- a) Machine Learning
 - b) Networking
 - c) Data Entry
 - d) File Management
7. **Which of the following is a limitation of AI?**
- a) High computational cost
 - b) Lack of common sense
 - c) Dependence on data
 - d) All of the above
8. **Which of the following is an ethical issue in AI?**
- a) Privacy
 - b) Bias
 - c) Accountability
 - d) All of the above
9. **Which of the following is a social concern related to AI?**
- a) Job displacement
 - b) Faster processing
 - c) Automation
 - d) Improved accuracy
10. **Which type of AI refers to a hypothetical system that would surpass human intelligence in all areas?**
- a) Narrow AI
 - b) General AI
 - c) Super AI
 - d) Expert System

Q. No.	Short Answer Question	Marks	CO	BTL
1	Define Artificial Intelligence (AI).	2	CO1	BTL 1 – Remember
2	Who coined the term Artificial Intelligence, and in which year?	2	CO1	BTL 1 – Remember
3	What is an intelligent system?	2	CO1	BTL 1 – Remember
4	List the main characteristics of intelligent systems.	2	CO1	BTL 1 – Remember
5	Explain the evolution of Artificial Intelligence briefly.	2	CO1	BTL 2 – Understand
6	Explain the role of learning and reasoning in an intelligent system.	2	CO1	BTL 2 – Understand
7	Differentiate between Narrow AI and General AI.	2	CO1	BTL 2 – Understand
8	Explain the different types of Artificial Intelligence with examples.	2	CO1	BTL 2 – Understand
9	Identify any four applications of AI in engineering.	2	CO1	BTL 1 – Remember
10	Explain how AI is used for predictive maintenance in engineering.	2	CO1	BTL 2 – Understand
11	What are the major limitations of Artificial Intelligence?	2	CO1	BTL 1 – Remember
12	Explain any two challenges involved in implementing AI systems.	2	CO1	BTL 2 – Understand
13	How does AI bias affect the decision-making process?	2	CO1	BTL 3 – Apply
14	Suggest suitable measures to address privacy issues in AI systems.	2	CO1	BTL 3 – Apply
15	Analyze the possible social impact of AI on employment and society.	2	CO1	BTL 4 – Analyze

Q. No.	Long Answer Question	Marks	CO	BTL
1	Explain the concept of Artificial Intelligence and discuss its objectives, scope, and significance in modern computing.	6	CO1	BTL 2 – Understand
2	Describe the major milestones in the development of Artificial Intelligence from its inception to the present era.	6	CO1	BTL 2 – Understand
3	Explain the important characteristics of an intelligent system. How do these characteristics distinguish an intelligent system from a conventional computer system?	6	CO1	BTL 4 – Analyze
4	Compare different types of Artificial Intelligence based on their capabilities and functionality. Provide suitable examples for each type.	6	CO1	BTL 4 – Analyze

5	Explain the role of Artificial Intelligence in modern engineering. Discuss its applications in design, manufacturing, automation, and maintenance.	6	CO1	BTL 2 – Understand
6	Analyze how Artificial Intelligence can be used to solve complex engineering problems. Illustrate your answer with suitable examples.	6	CO1	BTL 4 – Analyze
7	Discuss the major limitations of current Artificial Intelligence systems. Explain the issues related to data dependency, computational requirements, explainability, and reliability.	6	CO1	BTL 2 – Understand
8	Analyze the major challenges involved in developing reliable and intelligent AI systems for real-world applications.	6	CO1	BTL 4 – Analyze
9	Explain the ethical concerns in Artificial Intelligence with reference to fairness, privacy, bias, transparency, and responsible AI development.	6	CO1	BTL 2 – Understand
10	Analyze the impact of Artificial Intelligence on employment and the future of work. Discuss both opportunities and challenges.	6	CO1	BTL 4 – Analyze
11	Discuss the applications of Artificial Intelligence in healthcare, education, transportation, and finance. Explain how AI improves decision-making in these domains.	6	CO1	BTL 2 – Understand
12	Explain the role of Artificial Intelligence in automation and smart manufacturing. Discuss its benefits and limitations in engineering industries.	6	CO1	BTL 2 – Understand
13	Analyze the potential future developments in Artificial Intelligence and their impact on engineering and society.	6	CO1	BTL 4 – Analyze
14	Compare the advantages and limitations of Artificial Intelligence with traditional computing systems.	6	CO1	BTL 4 – Analyze
15	Evaluate the overall impact of Artificial Intelligence on society by considering its technological benefits, ethical concerns, employment effects, and future challenges.	6	CO1	BTL 5 – Evaluate

Viva Questions

1. What is Artificial Intelligence (AI)?
2. What is the main objective of Artificial Intelligence?
3. Who is known as the father of AI, and why is 1956 important in AI history?
4. What are the major characteristics of an intelligent system?
5. What are the different types of Artificial Intelligence?
6. What is Narrow AI? Give one example.
7. What is the difference between Artificial Intelligence and Machine Learning?
8. What is Generative AI? Give some examples.
9. What are the major applications of AI in engineering?
10. What is predictive maintenance, and how does AI help in it?
11. What are the major limitations of Artificial Intelligence?
12. Why is high-quality data important for AI systems?
13. What is AI bias? How can it affect decision-making?
14. What are the major ethical and social issues related to AI?
15. What is the future of Artificial Intelligence?

Web Reference

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